2025_Hexaware_Actual_Test_4

Test Summary

- No. of Sections: 2
- No. of Questions: 90
- Total Duration: 90 min

Section 1 - Aptitude

Section Summary

- No. of Questions: 60
- Duration: 60 min

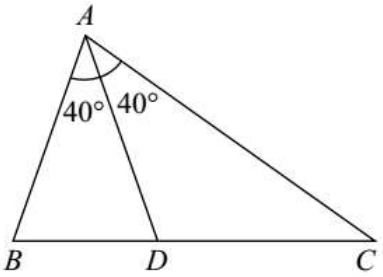
Additional Instructions:

None

Q1. The average age of a class of 22 students is 21 years. The average increases by 1 when the teacher's age is also include. What is the age of the teacher?

- 44 years
- 46 years
- 38 years
- 24 years

Q2. In the following figure, If BC = 8 cm, AB = 6 cm, AC 9 cm, then DC is equal to :



- 7 cm
- 4.8 cm
- 7.2 cm
- 4.5 cm

Q3. The length of a rectangular floor is more than its breadth by 200%. If Rs. 324 is required to paint the floor at the rate of Rs. 3 per sq m, then what would be the length of the floor?

- 27 m

18 m

24 m

28 m

Q4. The difference between the two numbers is 1365. When the larger number is divided by the smaller one, the quotient is 6 and the remainder is 15. The smaller number is:

240

270

295

360

Q5. The sum of the two numbers is 272 and their LCM is 960. Then their HCF is

16

8

32

64

Q6. What is the highest power of 5 that divides 90 x 80 x 70 x 60 x 50 x 40 x 30 x 20 x 10?

10

15

14

8

Q7. Convert the following number to base 10:
412 in base 6

152

1524

110011100

230

Q8. Jaya bought a car worth Rs.3,50,000 four years ago. If the value of the car depreciates over time at a fixed rate of 10% per annum, then what will be the present worth of the car?

Rs.2,55,150

Rs.2,30,000

Rs.2,50,000

Rs.2,29,635

Q9. In a group of 6 boys and 4 girls, four children are to be selected. In how many different ways can they be selected such that at least one boy should be there?

159

194

205

209

Q10. How many nine-digit numbers have an even sum of their digits?

9! – 6!

14×10^7

45×10^7

$10^9 / 2$

Q11. If two cards are drawn from a well-shuffled deck of 52 cards with replacement, then the probability that both are queen is

1/13 * 1/13

1/13 + 1/13

1/13 * 1/17

1/13 * 4/51

Q12. The difference in discounts between two successive discounts of 8% each and a single discount of 16% on Rs.2000 is

6.4

4.6

12.8

12.6

None of these

Q13. In an AP, $S_p = q$, $S_q = p$ and S denotes the sum of the first r terms. Then, S_{q+p} is equal to

0

-(p + q)

p + q

pq

Q14. In what proportion must wheat at Rs.1.60 per kg be mixed with wheat at Rs.1.45 per kg, so that the mixture is worth Rs.1.54 per kg?

2 : 3

3 : 2

3 : 4

4 : 3

Q15. The ratio of Science and Arts students in a college is 4:3. If 24 Science students shift to Arts then the ratio becomes 1:1, find the total strength of Science and Arts students.

280

252

308

336

Q16. A sum of money was invested at SI at a certain rate for 3 years. Had it been invested at a 4% higher rate, it would have fetched Rs. 480 more. Find the principal.

Rs. 4000

Rs. 4400

Rs. 5000

Rs. 3500

Q17. On a certain sum of money, compound interest earned at the end of three years is Rs. 1456. The compound interest at the end of two years is Rs. 880. Compute the principal invested.

Rs. 2,400

Rs. 2,800

Rs. 2,000

Rs. 1,600

Q18. If $a^2 + b^2 = c^2$, then $1/\log_{c+a} b + 1/\log_{c-a} b =$

1

2

-1

-2

Q19. A garrison of 1500 men is provisioned for 60 days. After 25 days the garrison is reinforced by 500 men. How long will the remaining provisions last?

24 days

21.75 days

26.25 days

52 days

Q20. To fill a cistern, pipes A, B, and C take 25 minutes, 15 minutes and 12 minutes respectively. Find the time in minutes that the three pipes together will take to fill the cistern(approximately).

5 minutes

10 minutes

15 minutes

25 minutes

Q21. **Fill in the blanks with a suitable preposition:**

He has no liking _____ cards.

for

with

on

in

Q22. In the question below is given a statement followed by three courses of action numbered I, II, and III. On the basis of the information in the statement, you have to assume everything in the statement to be true, then decide which of the three given suggested courses of action logically follows for pursuing. Then decide which of the answer (1), (2), (3), (4), and (5) is the correct answer.

Statement: The Department of Education has recommended that the primary-level admission to Government and Government aided schools, should be done purely by random selection and not by admission tests. This is necessitated as the number of admission seekers is much more than available seats.
The course of action :
I. The government should instruct the private schools also to follow the same practice.
II. The government should set up an independent body to regulate the primary level admissions.
III. The schools should be asked to select students only from neighboring areas of the school.

None follows

Only II and III follows

Only I and II follow

Only II follows

Q23. **A number of sentences are given below which, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a letter. Choose the most logical order of sentences from among the four given choices to construct a coherent paragraph.**

A. It enables a selector to interview a candidate sitting miles away, sometimes in a different country, from his chamber, thereby saving him valuable time in making a selection.
B. Hence while this methodology is economical, it cannot replace the personal interview completely.
C. Telephonic interviews have some unique advantages and disadvantages for both the candidate and the selector.
D. Personal interviews are making way for, interviews conducted over the telephone, thanks to the advent of new technologies and improved communication systems.
E. One common disadvantage both the candidate and the selector face is that one can only impress through words and voice, and cannot create a positive image through dress and body language.

- DCAEB
- CAEBD
- EDACB
- CBDEA

Q24. **Fill in the blanks:**

_____dog over there is mine.

- there
- that
- this
- these

Q25. **Find the error in the underlined sentence and answer the correct one from the given options.**

The granddaddy of manufacturers, the Shutters Company, headquartered in Milton, is nearly 50 years <u>as old as any of their supposed</u> predecessors.

- as old as any of their supposed
- older than any of their supposed
- as old as their supposed
- older than any of their supposedly

Q26. **Convert from Direct to Indirect speech.**

She said, "I saw a snake here."

- She said that she saw a snake here.
- She said that she saw a snake there.

She said that she had seen a snake here.

She said that she had seen a snake there.

Q27. **Choose the word opposite in meaning to the given word**

Mutinous

Abstruse

Esoteric

Recondite

Compliant

Q28. **Out of the four alternatives choose the one which can be substituted for the given words/sentence.**

A plan of a journey, including the route and the places that you visit.

Iteration

Itinerary

Signage

Travelogue

Q29. **Identify the one that is opposite in meaning (antonym) to the question word and mark.**

Bucolic

lecherous

rustic

fragrant

urban

Q30. **From the given options choose the one that best expresses the given sentence in active/passive voice:**

The machine wraps the bread automatically.

The bread is being wrapped automatically.

The bread was wrapped automatically

The bread have been wrapped automatically

The bread is wrapped automatically

Q31. If the fifth and 12th letters of the word 'GLORIFICATIONS' are interchanged, also fourth and 14th letters, third and 10th, second and 11th, first and 13th then which of the following would be 12th letter from the right?

T

O

R

I

Q32. Pointing to a man in a photograph, Reena said "His mother's only daughter is my mother." How is Reena related to that man?

Nephew

Sister

Niece

Wife

Q33. What was the day on 26th Jan 1950, when 1st Republic Day of India was celebrated?

Monday

Tuesday

Thursday

Friday

Q34. Find the day of the week on 25th December 1995.

Friday

Saturday

Sunday

Monday

Q35. In a certain code language, if ANIMAL is written as AAMINL, then the word PLANT will be written as

PNALT

PANTL

PTLAN

PANLT

Q36. Two statements numbered I and II are given below. You have to decide whether the data provided in the statements are sufficient to answer the question.

What is the code for 'juicy apple'?

Statements:
I. The code for 'apple juice is cedar' is 'ka la do ni', the code for 'cedar vinegar is good' is 'do re la mi' and the code for 'buy some juicy apples' is 'vu hi wo pe'.
II. The code for 'good orange and apple' is 'ni xe co mi', the code for 'apple juice is fresh' is 'ka la ni po' and the code for 'fresh and juicy orange' is 'xe co pe po'.

If the data in statement I alone are sufficient to answer the question while data in statement II alone is not sufficient.

If the data in statement II alone is sufficient to answer the question while the data in statement I alone is not sufficient.

If the data in both statements I and II together are not sufficient to answer the question

If the data either in statement I alone or in statement II alone is sufficient to answer the question.

If the data in both statements I and II together are necessary to answer the question

Q37. The time is 9 am., when can Utham get the next bus to Vijainagar from Mirpur?

Statement I: Buses for Vijainagar leave after every 60 minutes, till 11 am.
Statement II: Half an hour ago, a bus left for Vijainagar.

If the data in statement I alone is sufficient to answer

If the data in statement II alone is sufficient to answer

If the data either in I or II alone is sufficient to answer

If the data even in both the statements together are not sufficient to answer

If the data in both the statements together are needed to answer

Q38. Rana travels 10 km to the north, turns left and travels 4 km, and then again turns right and covers another 5 km, and then turns right and travels another 4 km. How far is he from the right starting point?

15 km

4 km

5 km

10 km

Q39. **Considering given statements as true, select a logical conclusion based on the given statements.**

Statements:
1. Some roads are highways.
2. Some highways are seaways.
3. Some seaways are airways.

Conclusions:
I. Some airways are roads.
II. Some seaways are roads.
III. Some airways are highways.
IV. Some roads are airways.

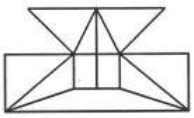
Only III and IV follow

Only I and II follow

All of the conclusions follow

None of the conclusions follow

Q40. Find the minimum number of straight lines required to make the given figure.



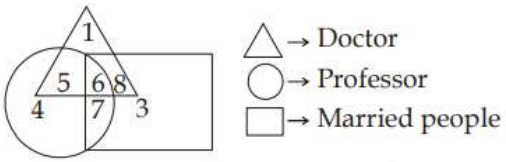
16

17

18

19

Q41. Which number indicates doctors who are not married



5

1

6

8

Q42. A four-centimeter cube has been painted yellow on all its sides. It is cut into one-centimeter cubes. How many cubes will be there with only one side painted yellow?

8

16

24

32

Common Content:

Management is a set of processes that can keep a complicated system of people and technology ___1___smoothly. The most important aspects of management include planning, budgeting, organizing, staffing, controlling, and problem-solving. Leadership is a set of processes that creates organizations in the first place or _____2___ them to significantly changing circumstances. Leadership defines what the future should look like, ___3___ people with that vision, and inspires them to make it happen despite the obstacles. This _____4___ is absolutely crucial for our purposes here: Successful _____5____ is to 90 percent leadership and only 10 to 30 percent management.

Q43. Choose the appropriate word for the blank numbered (2)

qualify

reconcile

adapts

dislocate

Q44. Choose the appropriate word for the blank numbered (1)

dawdling

running

appreciating

enduring

Q45. Choose the appropriate word for the blank numbered (5)

realignment

transition

stagnation

stoppage

Q46. Choose the appropriate word for the blank numbered (3)

enlists

disregards

aligns

ally

Q47. Choose the appropriate word for the blank numbered (4)

accord

acumen

distinction

estimation

Common Content:

Read the passage and answer the question given below.

In choosing a method for determining climatic condition that existed in the past, paleo-climatologists invoke four principal criteria. First, the material-rocks, lakes, vegetation, etc on which the method relies must be widespread enough to provide plenty of information, since analysis of material that is rarely encountered will not permit correlation with other regions or with other periods of geological history. Second, in the process of formation, the material must have received an environmental signal .that reflects a change in climate and that can be deciphered by modern physical or chemical means. Third, at least some of the material must have retained the signal unaffected by subsequent changes in the environment. Fourth, it must be possible to determine the time at which the inferred climatic conditions held. This last criterion is more easily met in dating marine sediments, because dating of only a small number of layers in a marine sequence allows the age of other layers to be estimated fairly reliably by extrapolation and interpolation. By contrast, because sedimentation is much less continuous in continental regions, estimating the age of a continental bed from the known ages of beds above and below is more risky.

One very old method used in the investigation of past climatic conditions involves the measurement of water levels in ancient lakes. In temperate regions, there are enough lakes for correlations between them to give us a reliable picture. In arid and semiarid regions, on the other hand, the small number of lakes and the great distances between them reduce the possibilities for correlation. Moreover, since lake levels are controlled by rates of evaporation as well as by precipitation, the interpretation of such levels is ambiguous. For instance, the fact that lake levels in the semiarid southwestern United States appear to have been higher during the last ice age than they are now was at one time attributed to increased precipitation. On the basis of snow-line elevations, however, it has been concluded that the climate then was not necessarily wetter than it is now, but rather that both summers and winters were cooler, resulting in reduced evaporation.

Another problematic method is to reconstruct former climates on the basis of pollen profiles. The type of vegetation in a specific region is determined by identifying and counting the various pollen grains found there.

Although the relationship between vegetation and climate is not as direct as the relationship between climate and lake levels, the method often work well in the temperature zones. In arid and semiarid regions in which is not much vegetation however, small changes one or a few plant types, can change the picture dramatically, making accurate correlations between neighbouring areas difficult to obtain.

Q48. It can be inferred from the passage that an environmental signal found in geological material would not be useful to paleo-climatoiogists if it _____.

- Reflected a change in climate rather than a long-term climatic condition.
- Was incorporated into a material as the material was forming
- Also reflected subsequent environmental changes.
- Was contained in a continental rather than a marine sequence

Q49. Which of the following statements about the difference between marine and continental sedimentation is supported by information in the passage?

- Data provided by dating marine sedimentation is more consistent with researchers' findings in other disciplines than is data provided by dating continental sedimentation.
- It is easier to estimate the age of a layer in a sequence of continental sedimentation than it is to estimate the age of a layer in a sequence of marine sedimentation.
- Researchers are more often forced to rely on extrapolation when dating a layer of marine sedimentation than when dating a layer of continental sedimentation.
- Marine sedimentation is much more continuous than is continental sedimentation.

Q50. It can be inferred from the passage that paleo-climatologists have concluded which of the following on the basis of their study of snow-line elevations in the southwestern United States?

- There is usually more precipitation during an ice age because of increased amounts of evaporation.
- Lake levels in the semiarid southwestern United States were lower during the last ice age than they are today.

During the last ice age, cooler weather led to lower lake levels than paleo-climatologists had previously assumed.

The high lake levels during the last ice age may have been a result of less evaporation rather than more precipitation.

Common Content:

Study the following information carefully and answer the given questions:
The following table shows the total number of Professionals from different departments in different years in a certain city and the percentage of females among them.

Years	Engineering		Law		Medicine	
	Total employees	% of female	Total employees	% of female	Total employees	% of female
2013	650	48 %	450	46 %	180	35 %
2014	720	45 %	560	35 %	160	45 %
2015	680	60 %	620	55 %	150	38 %
2016	540	40 %	650	56 %	200	42 %
2017	660	50 %	700	40 %	120	55 %

Q51. Number of medical professionals in the year 2014 is approximately what percentage the number of engineering male professionals in the year 2016?

42 %

49 %

28 %

15 %

Q52. Find the ratio between the number of male medical professionals in the year 2014 to that of the number of female engineering professionals in the year 2013.

23: 57

51: 79

5: 13

11: 39

Q53. Find the average number of male law professionals in the year 2013, 2015 and 2017 together?

314

352

287

275

Q54. What is the difference between the total number of engineering professionals and total number of medical professionals in all the given years together?

2450

2300

2440

3340

Q55. Find the difference between the number of male engineering professionals in the year 2015 to that of the number of female Law professionals in the year 2013.

43

24

65

52

Common Content:

Read the given information carefully and answer the questions that follow.
Eleven friends M, N, O, P, Q, R, S, T, U, V and W are sitting in the first row of the stadium watching a cricket match.
T is to the immediate left of P and third to the right of U. V is the immediate neighbour of M and N and third to the left of S. M is the second to the right of Q, who is at one of the ends. R is sitting next to the right of P and P is second to the right of O.

Q56. Who is sitting in the center of the row?

N

S

O

U

Q57. Which of the following people are sitting to the right of S?

OTPQ

OTPR

UNVM

UOTPR

Q58. Which of the following statements is true with respect to the above arrangement?

There are three persons sitting between P and S

W is between M and V

N is sitting between V and U

S and O are neighbours sitting to the immediate right of T

Q59. Which of the following people are sitting to the left of V?

QWM

NUS

TPR

USO

Q60. Which of the following statements is true with respect to the above arrangement?

U is between N and S

There are two people right to O.

Q is immediate to right of W.

V is right to N.

Section 2 - Pseudocode and Computer Fundamentals

Section Summary

- No. of Questions: 30
- Duration: 30 min

Additional Instructions:

None

Q1. What will be the output of the following pseudocode?

[Note:- && : Logical AND - The logical AND operator (&&) returns the boolean value True (or 1) if either or both operands is True and returns False(or 0)
If (X) gets executed if the value inside if( ), ie, x is not zero].

```
1  integer a, b, c
2  set a=11, b= 21
3  for(each c from 16 to 17)
4  if ( c-10 > 8)
5  if (1 && 1)
6  a= a + c
```

```
6   a= a + c
7   end if
8   a= a + 1
9   else
10  a= 0
11  end if
12  end for
13  print b + a
```

- 11
- 14
- 21
- 50

Q2. What will be the output of the following pseudocode?

[Note:-&& : Logical AND - The logical AND operator (&&) returns the boolean value True (or 1) if either or both operands is True and returns False(or 0)].

```
1  integer a, b
2  set b= 2
3  set a= b + 3
4  if (a && b - a && 1)
5  a=a +1
6  end if
7  print a
```

- 8
- 5
- 6
- 4

Q3. What will be the output of the following pseudocode?

```
1  integer n, c
2  set n= 128, c= 4
3  while( n NOT EQUALS 0)
4  n= n / 10
5  c = c - 1
6  end while
7  print c
```

- 0
- -1
- 2

1

Q4. What will be the output of the following pseudocode?

```
integer a[5]
set a[5]= { 1,2,3,4}
integer m, j
set m= 1
for ( each j from 0 to 3)
m= m + a[j] + a[ j] + 6
end for
print m
```

43

45

23

46

Q5. How many times the following pseudocodes will print hello?

```
integer k, p, j
set k= 1, p= 4
int j= p + k / 2
if ( j equals p)
print "hello"
end if
j= p + k / 5
if ( j not equals p)
print " hello "
end if
print " hello "
```

Three times

One times

Two times

will not print hello at all

Q6. What will be the output of the following pseudocode?

[Note: If (X) gets executed if the value inside if( ), ie, x is not zero].

```
integer k, p
set k= 56
p = k + 8 - 78 / 8 * 4/7 - 5/ 2
if (p )
print "hello "
else
print " bye"
end if
```

None of these mentioned options

hello

hello bye

bye

Q7. What will be the output of the following pseudocode?

[Note:-&& : Logical AND - The logical AND operator (&&) returns the boolean value True (or 1) if either or both operands is True and returns False(or 0)].

```
1  integer a, b, c
2  set a= 1, b= 2
3  for (each c from 216 to 217)
4  if( c - 200 > 100)
5  if( 1 && 1)
6  a= a + c
7  end if
8  a= a+ c
9  else
10 a=0
11 end if
12 end for
13 print b + a
```

8

4

2

7

Q8. What will be the output of the following pseudocode?

```
1  integer a, b, c, d
2  set d= 6
3  for( each a from 1 to 2)
4  for(each b from a to 2)
5  for (each c from b to 2)
6  print d
7  end for
8  end for
9  end for
```

6 6 6 6 6 6

6

6 6 6

6 6 6 6

Q9. What will be the output of the following pseudocode?

[Note: MOD finds the remainder after the division of one number by another. For example, the expression "5 MOD" would evaluate to 1 because 2 leaves a quotient of 2 and a remainder 1].

```
1  integer a, b, c
2  set a= 108, b= 2, c= 3
3  if (( a MOD 9) EQUALS 0)
4  b= b + (a MOD 2)
5  print b
6  else
7  c= c + ( a MOD 7)
8  print c
9  end if
```

- 2
- 7
- 1
- 3

Q10. What will be the output of the following pseudocode for a=6?

```
1  integer fun (integer a)
2  if (a EQUALS 0)
3  return a+ 9
4  else
5  return fun(a-1)
6  end function fun()
```

- 6
- 0
- 9
- 1

Q11. What will be the output of the following pseudocode?

[Note: MOD finds the remainder after the division of one number by another. For example, the expression "5 MOD" would evaluate to 1 because 2 leaves a quotient of 2 and a remainder 1
| | : Logical OR - The logical OR operator (| |) returns the boolean value True (or 1) if either or both operands is True and returns False(or 0), Otherwise].

```
1  integer i
2  set i=45
3  if( i mod 2 EQUALS 0 OR i mod 2 AND i)
4  print i mod 2
5  else
6  print i
7  end if
```

- 1

0

2

45

Q12. Consider the following algorithm:

p is a pointer to the first node of the linked list, Which of the following operations is performed by the function fun?

```
1  integer fun (Node *{p)
2  integer x= MIN_INT
3  while(p not equals NULL)
4  if (p->data < x)
5  return 0
6  else
7  x= p -> data
8  p -> next
9  end if
10 end loop
11 return x
12 end fun
```

function fun detects a loop in the linked list, it return 1 when a loop is found and 0 when no loop is found

None of the mentioned options

Functions fun finds the largest element of the linked list

Function fun checks whether a linked lists is sorted or not. It returns 0 when the list is not sorted and 1 when it is sorted

Q13. What will be the output of the following pseudocode?

```
1  integer a
2  string str1
3  set str1= " momo"
4  for (each a from 1 to 2)
5  str1= str1 + " mm"
6  end for
7  print (stringLength(str1))
```

7

4

6

11

Q14. What will be the output of the following pseudocode?

```
1  Integer value, n
2  Set value = 1, n = 45
3  while(value less than equal to n)
4    value = value << 1
```

```
5 end loop
6 Print value
```

- 16
- 32
- 64
- 36

Q15. How many times the following pseudocode will print "btech".

```
1 integer a, b
2 for(each a from 0 to 3)
3 for(each b from 0 to a)
4 print "btech"
5 end for
6 end for
```

- 3 times
- 9 times
- 10 times
- 11 times

Q16. Which of the following is used for Computer Games?

- Mouse
- Trackpad
- Joystick
- Touchpad

Q17. Which of the following can be attached with an email?

1. Word document
2. Images
3. PowerPoint Presentation
4. Audio/Video files

Choose the correct answer from the options given below

Only 1

All 1, 2, 3 and 4

Only 1, 2 and 3

Only 1 and 2

Q18. 1 megabyte(MB) = _____ kilobytes(KB) in binary.

256

100

1024

1048

Q19. In networking, which of the following is defined as a firewall?

The physical boundary of a computer network

A system designed to prevent unauthorized access

A web browsing software

An operating system of a computer network

Q20. Operating system is an example of which of the following types of software?

Application software

System software

Utility software

None of the mentioned options

Q21. Which of the following text functions in MS Excel is case-sensitive?

1. FIND
2. EXACT
3. LEN

2 & 3

1 & 3

1 & 2

1, 2 & 3

Q22. Outlook Express is a/an:

Search engine

Website

Browser

E-mail client

Q23. If data need to be added or deleted frequently from a file, which among the following should be done?

Indirect sequential

Sequential

Index sequential

Direct

Q24. Which of the following options is the largest unit of storage in the computer system?

Kilobyte

Terabyte

Gigabyte

Megabyte

Q25. Which of the following options should a person choose if he has to mail every person present in on ongoing mail thread?

Reply

New mail

Forward

Reply all

Q26. A block of text added to a mail automatically at its end is called a/an:

Attachment

Footer

Signature

Copyright text

Q27. Find the odd one out.

External hard drive

Pen drive

Memory card

RAM

Q28. Which of the following can be added to the signature of the email?

1. Text
2. Link
3. Image

All 1, 2 and 3

Only 2 and 3

Only 1 and 3

Only 1

Q29. Which of the following is a correct format of email address?

- name@website@info
- name.website.com
- www.nameofebsite.com
- name@website.info

Q30. Which of the following statements is correct for the Incognito mode of the browser?

1. It won't save your browsing history
2. It won't keep the bookmarks you create
3. It won't hide your activities from websites you visit

- 2 & 3
- 1, 2 & 3
- 1 & 2
- 1 & 3

Answer Key & Solution

Section 1 - Aptitude

Q1

44 years

Solution

We have, Average x Number of = Total

21 years x 22 Numbers = 462 years .... (1)

and 22 years x 23 Numbers = 506 years .... (2)

Equation (1) gives the sum total of ages of all the students while equation (2) gives the sum total of ages of all the students along with the teacher.

Teacher's age = (2) – (1) = 506 – 462 = 44 years.

Q2

4.8 cm

Solution

From above given figure , BC = 8 cm, AB = 6 cm and AC 9 cm

Using the sine fomula, we have

a/SinA = b/sinB = c/sinC

∴ sin40°/BD = sin(40°+C)/6

⇒ sin40°/(B - DC) = sin[180-(40°+C)]/6 = sin(140 - C)]/6 ..... (i)

Also, sin40°/DC = sin(140 - C)]/9 ..... (ii)

∴ From (i) and (ii) given

DC/(8 - DC) = 9/6 = 3/2

⇒ 2DC = 24 - 3DC ⇒ 5DC = 24

⇒ DC = 24/5 ⇒ DC = 4.8

Q3

18 m

Solution

Let the length and the breadth of the floor be l m and b m respectively.

l = b + 200% of b = b + 2b = 3b

Area of the floor = 324/3 = 108 sq m

l b = 108 i.e., l * l/3 = 108

$l^2 = 324 \Rightarrow l = 18m.$

Q4

270

Solution

Let the smaller number be x. Then larger number = (x + 1365).

$x + 1365 = 6x + 15$

$5x = 1350$

$x = 270$

Smaller number = 270.

Q5

16

Solution

We know that the addition of two numbers is divisible by their HCF.

So, the HCF of the numbers divides the sum 272.

$272 = 2 \times 2 \times 2 \times 2 \times 17$

$960 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 5$

$HCF = 2 \times 2 \times 2 \times 2 = 16$

Q6

10

Solution

Highest power of 5 is

90= 5*18

80= 5* 16

70= 5*14

60= 5*12

$50=5^2*2$

40=5*8

30=5*6

20=5*4

10=5*2

There are 10 fives

Q7

152

Solution

$(4 \times 36) + (1 \times 6) + (2 \times 1) = 152$

Q8

Rs.2,29,635

Solution

Each year decreases by 10% and continuous 4 years = 350000*(90/100)*(90/100)*(90/100)*(90/100)=229635

Q9

209

Solution

We may have (1 boy and 3 girls) or (2 boys and 2 girls) or (3 boys and 1 girl) or (4 boys).

∴ Required number

of ways= $({}^6C_1 \times {}^4C_3) + ({}^6C_2 \times {}^4C_2) + ({}^6C_3 \times {}^4C_1) + ({}^6C_4)$=

$({}^6C_1 \times {}^4C_1) + ({}^6C_2 \times {}^4C_2) + ({}^6C_3 \times {}^4C_1) + ({}^6C_2) = (6 \times 4) + \left(6 \times \frac{5}{2 \times 1} \times 4 \times \frac{3}{2 \times 1}\right) + \left(6 \times 5 \times \frac{4}{3 \times 2 \times 1} \times 4\right)$

= (24 + 90 + 80 + 15)= 209.

Q10

45×10^7

Solution

The first eight digit can be chosen arbitrarily. There are 9×10^7 ways to do this.

Then the last digit can always be chosen in exactly 5 ways.

(If the sum of the previous eight digits is odd, then we must choose an odd digit. Otherwise the last digit must be even)

Hence the total ways = $9 \times 10^7 \times 5 = 45 \times 10^7$.

Q11

1/13 * 1/13

Solution

Total cards=52

Both are queen=4/52 * 4/52 = 1/13 * 1/13

Q12

12.8

Solution

Two successive discounts of 8% = [-8-8+(8*8/100)]% = -15.36%

Diff = 16 – 15.36 = 0.64%

Required diff = 2000×(0.64/100) = Rs. 12.8

Q13

-(p + q)

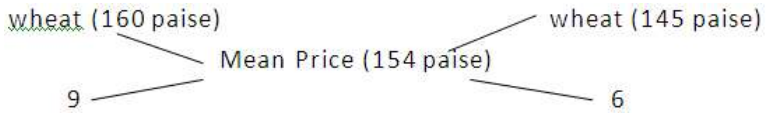
Solution

No Solution

Q14

3 : 2

Solution



(Dearer wheat) : (Cheaper wheat) = 9 : 6 = 3 : 2.

Q15

336

Solution

Let the number of Science and Arts students be 4x and 3x respectively.

4x – 24 and 3x + 24 are in the ratio 1 : 1 (they are equal)

4x – 24 = 3x + 24

x = 48.

The total strength is 4(48) + 3(48) = 336.

Q16

Rs. 4000

Solution

Excess money obtained = 3 years @ 4% per annum

= 12% of Principal

So, according to the question, 12% = Rs. 480

So, 100% = Rs. 4000

Q17

Rs. 2,000

Solution

Let principal = P, rate of interest = r %

CI earned at the end of three years = $P(1 + r)^3 - P = 1456$

$=> P(3r^2 + 3r + r^3) = 1456$

CI earned at the end of two years = $P(1 + r)^2 - P = 880$

$=> P(r^2 + 2r) = 880$

Dividing one by the other we get: $3r^2 +3r +r^3/(r^2 +2r) = 1456/880$

We can cancel 'r' and solve the resulting quadratic. However, let us see if we can spot something in the numbers.

1456/880 = 728/440

Remember that the amounts at the end of three years and two years are linked to $(1 + r)^3$ and $(1 + r)^2$ respectively.

Now observe that $1728 = 12^3$ and 144 is 12^2.

So, 728/440 = 1728-1000/(1440-1000) = 1.728-1/(1.440-1)

=>r = 20%

Alternatively:

$(r^2 + 3r + 3) \times 440 = (r + 2) \times 728$

$(r^2 + 3r + 3) \times 55 = (r + 2) \times 91$

$55r^2 + 165r + 165 = 91r + 182$

$55r^2 + 74r -17 = 0$

$55r^2 + 85r - 11r - 17 = 0$

$5r (11r + 17) -1 (11r + 17) = 0$

r = 0.2 or a negative number. Or, r has to be 20%.

The question is " Compute the principal invested."

Rs. 2,000 is the principal amount.

Hence, the answer is Rs. 2000.

Q18

2

Solution

$1/\log_{c + a} b + 1/\log_{c-a} b = \log_b(c+a) + \log_b(c-a)$

$=\log_b(c^2-a^2) = \log_b b^2 = 2.$

Q19

26.25 days

Solution

Since the garrison is reinforced by 500 men therefore there are (1500 + 500) or 2000 men

now; since 60 – 25 = 35 days

=> The provisions left would last 1500 men 35 days.

Provisions left would last 1 man 35 × 1500 days

=> Provision left would .last 2000 men for 35 × 1500/2000 = 26.25 days.

Short-cut:

By work equivalence method 1500 × 60 = (1500 × 25) + (2000 × X)

Solve to get X = 26.25 days.

Q20 5 minutes

Solution

In 1 min, A can fill 1/25, B 1/15 and C 1/12 of the cistern

In 1 min together they can fill 1/25 + 1/15 + 1/12

= (12 + 20 + 25)/300

= 57/300 of the cistern

So, they take 300/57 = 5.26 ~ 5 min to fill the cistern together

Q21 for

Solution

Liking usage :

Liking for something / somebody . In this case it is Liking for cards(something).

Q22 Only II follows

Solution

Mere instructions to private schools regarding admission procedure will not work. Its implementation to a larger extent would be possible if an independent body to regulate the primary level admission is set up. Ans 4

Q23 DCAEB

Solution

Statement 'D' is the apt choice to begin the paragraph because only 'D' gives an introduction to the subsequent sentences as it talks of 'the advent of new technologies and improved communication system' in relation to the 'interviews over phone'.

Hence, Option (1) DCAEB is the correct answer

Q24 that

Solution

There is used for things far away. So the answer should be 'that'

Q25 older than any of their supposed

Solution

The expression 'as old as' indicates equality of age, but the sentence indicates that the Shutters company predates other companies. Older than makes the point of comparison clear. The adjective 'supposed' should be used rather than the adverb 'supposedly'.

Q26 She said that she had seen a snake there.

Solution

The given sentence is in direct speech and we will change it into indirect speech. The inverted commas (" ") used in Direct Narration will be removed in Indirect Narration and "that" conjunction will be used. The tense of the reported verb is in simple past tense and according to the rules of narration, it will be changed to past perfect tense.

"She said that she had seen a snake there" will be the correct conversion.

Q27 Compliant

Solution

Mutinous means refusing to obey the orders of a person in authority.

Abstruse - difficult to understand; obscure.

Esoteric - intended for or likely to be understood by only a small number of people with a specialized knowledge or interest.

Recondite - (of a subject or knowledge) little known; abstruse.

Compliant - disposed to agree with others or obey rules, especially to an excessive degree; acquiescent.

Q28 Itinerary

Solution

Iteration-repetition of a process

Itinerary-planned route or travel plans

Signage-signs collectively

Travelogue-a film, book, or illustrated lecture about the places visited by or experiences of a traveler

Thus the right answer is itinerary

Q29 urban

Solution

'Bucolic' means a country person. Hence, 'Urban' is opposite in meaning to the word 'bucolic'.

Q30 The bread is being wrapped automatically.

Solution

Solution:

The given sentence is in simple tense and active voice.

When the voice is changed, Wraps becomes is being wrapped. The right answer is"The bread is being wrapped automatically."

Q31 T

Solution

By interchanging the letters according to the question, the following word is formed:

Given: G L O R I F I C A T I O N S

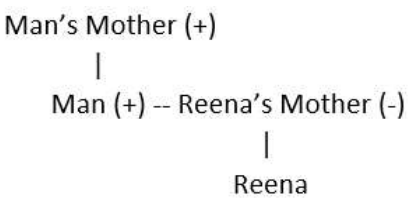
Interchanged: N I T S O F I C A O L I G R

12th letter from the right is T

Q32 Niece

Solution

Male(+), Female(-), Couple(=), Siblings(--), Next Generation(|)



so, Reena is the Man's Niece.

Q33 Thursday

Solution

The number of odd days in first 1600 years are 0

number of odd days in 300 years = 1

number of odd days in 49 years = (12 * 2 + 37 * 1) = 61 days = 5 odd days

total number of odd days in 1949 years = 1 + 5 = 6 odd days

Now look at the year 1950

jan 26 = 26 days = 3 weeks + 5 days = 5 odd days

Total number of odd days = 6 + 5 = 11 => 4 odd days

Therefore,Jan 26th 1950 was Thursday

Q34 Monday

Solution

300 years have 1 odd day.

94 years = (23 leap years + 71 ordinary years)

= (46 + 71) odd days = 117 odd days = 5 odd days.

Number of days from 1st Jan, 1995 to 25 th Dec, 1995 = 359 days.

= (51 weeks + 2 days) = 2 odd days.

:. Total number of odd days = (0 + 1 + 5 + 2) odd days = 1 odd day.

:. Required day is 'Monday'.

Q35 PNALT

Solution

PNALT

Q36

If the data in statement II alone is sufficient to answer the question while the data in statement I alone is not sufficient.

Solution

From I alone:

apple juice **is cedar** == ka **la do** ni

cedar vinegar **is** good == **do** re **la** mi

buy some juicy apples == vu hi wo pe

So, we get:

Is/cedar == la/do (not necessarily in the same order)

Apple/juice ==ka/ni (not necessarily in the same order)

Vinegar/good == re/mi (not necessarily in the same order)

So, the code for 'juicy apple' cannot be determined.

So, statement I alone is not sufficient.

From II alone:

good *orange and* **apple** == **ni** *xe co* mi

apple juice is **fresh** == ka la **ni** **po**

fresh *and* juicy *orange* == xe co pe **po**

so, we get

Apple == ni

And/ orange == xe/co (not necessarily in the same order)

Fresh == po

so, juicy == pe

So, the code for 'juicy apple' is 'ni pe'.

So, statement II alone is sufficient.

Q37

If the data in both the statements together are needed to answer

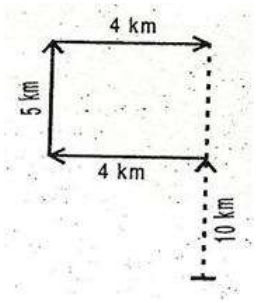
Solution

II reveals that the previous bus had left 8:30am, As given in I, the next bus would leave after 60 minutes i.e., 9:30 am

Q38

15 km

Solution



From the starting point he is 15km(10 + 5).

Ans.(1)

Q39

None of the conclusions follow

Solution

No Solution

Q40

17

Solution

No Solution

Q41

6

Solution

6, the doctor who are not married (5 + 1)

Q42

24

Solution

We know that Cubes with one surface painted can be find using 6 x $(X-2)^2$, where X is number of cuttings.

Here x = 4.

$\therefore$ 6 x $(4-2)^2$ = 24

Q43. adapts

Solution

The word "changing circumstances" after the second blank indicates that "adapts" will be the suitable answer.

Q44. running

Solution

A process will 'run' smoothly but not appreciate or endure.

Q45. transition

Solution

Among the given options, "Transition" is more suitable as it deals with transforming from one structure to another

Q46. aligns

Solution

Based on the plans for the future, one has to change their mindset/vision or "align" themselves accordingly.

Q47. distinction

Solution

The plan mentioned before the 4th blank, "distinction" is more suitable for the blank

Q48. Also reflected subsequent environmental changes.

Solution

o Solution

Q49. Data provided by dating marine sedimentation is more consistent with researchers' findings in other disciplines than is data provided by dating continental sedimentation.

Solution

No Solution

Q50. The high lake levels during the last ice age may have been a result of less evaporation rather than more precipitation.

Solution

No Solution

Q51. 49 %

Solution

Number of medical professionals in the year 2014 = > 160

The number of engineering male professionals in the year 2016 = > 540*(60/100) = > 324

Required % = 160/324 * 100 = 49%

Q52. 11: 39

Solution

The number of male medical professionals in the year 2014 = > 160*(55/100) = > 88

The number of female engineering professional in the year 2013 = > 650*(48/100) = > 312

Required ratio = 88: 312 = 11: 39

Q53. 314

Solution

The total number of male law professionals in the year 2013, 2015 and 2017 together = > 450*(54/100) + 620*(45/100) + 700*(60/100) = > 243 + 279 + 420 = 942

Required average = 942/3 = 314

Q54. 2440

Solution

otal number of engineering professionals in all the given years together = > 650 + 720 +

80 + 540 + 660 = 3250

otal number of medical professionals in all the given years together = > 180 + 160 + 150

200 + 120 = 810

equired difference = 3250 – 810 = 2440

Q55. 65

Solution

The number of male engineering professionals in the year 2015 = > 680*(40/100) = > 272

The number of female Law professionals in the year 2013 = > 450*(46/100) = > 207

Required difference = 272-207 = 65

Q56. U

Solution

Q	W	M	V	N	U	S	O	T	P	R

U is sitting in the center.

Q57. OTPR

Solution

Q	W	M	V	N	U	S	O	T	P	R

OTPR are sintting to the right of S

Q58. N is sitting between V and U

Solution

No Solution

Q59. QWM

Solution

Q	W	M	V	N	U	S	O	T	P	R

QWM are sitting left to V

Q60. U is between N and S

lution

o Solution

Section 2 - Pseudocode and Computer Fundamentals

Q1 21

Solution

No Solution

Q2 6

Solution

Initially b= 2 and a= 2 + 3 (a=5)

5 && 2 =0 , 5 && 1 = 1

If block executed for all the values except 0. Here if condition expression result value is -1 (which is considered as true) and increments 'a' value.

As a result, 6 gets printed.

Q3 1

Solution

The logic of the program is to find a number of digits in the number. The total digit for the given number 128 is 3, so 'c' decrements three times (4-3 = 1)

As a result, 1 gets printed.

Q4 45

Solution

Initially a[5]= { 1,2,3,4}

Inside loop all the array element are added twice and added again with 6 . Final result add with 1 (m value)

i.e a={1+1+6+2+2+6+3+3+6+4+4+6} + 1 (m value)

As a result, 45 gets printed.

Q5 Two times

Solution

Initially k= 1, p= 4

j= 4 + 0 = 4

if block condition is true so hello prints once and j value updated to (j= 4 + 0 = 4) .

Second if block condition is gets false and print outside hello.

So twice hello gets printed.

Q6

hello

Solution

i value set as 57 (shown below) , so if block gets executed and prints "hello"

56 + 8 - 78 / 8 * 4/7 - 5/ 2
56 + 8 - 9* 4/7 - 5/ 2
56 + 8 - 36/7 - 5/ 2
56 + 8 - 5 - 5/ 2
56 + 8 - 5 - 2
56 + 3- 2
56 + 1
57

Q7

2

Solution

Initially a= 1, b= 2

Else block gets executed assign a=0.

As a result 0+2 = 2 gets printed.

Q8

6 6 6 6

Solution

'd' value gets printed for below a,b and c values. (i.e) (a,b,c)

(1,1,1) , (1, 1, 2), (1, 2, 2), (2, 2,2) (totally 4 times 'd' value gets printed.

Q9

2

Solution

Initially a= 108, b= 2, c= 3

If block gets executed and update b value to 2 and b gets printed i.e 2.

Q10

9

Solution

In this recursive program, the recursive call is made until 'a' value becomes 0. fun(0) function call returns 9 and that value is passed to the first function call and 9 gets printed.

Q11

1

Solution

Initially i=45

If condition becomes true

(i.e) 45 %2 ==0 (False)

45%2 = 1(True)

i=45 (Except 0 all values are considered as True)

Therefore False OR True And True => True

Then 45%2=1 gets printed.

Q12

Functions fun finds the largest element of the linked list

Solution

While loop runs until pointer value becomes null (i.e) last value in the linked list.

When x is less than the current node value then update x with the current node value. Finally, x holds the maximum value in the linked list.

Q13

11

Solution

No Solution

Q14

64

Solution

No Solution

Q15

10 times

Solution

"btech" gets printed for below a,b values.

(i.e) (0, 0), (1, 0) , (1, 1), (2, 0) , (2, 1), (2, 2), (3, 0), (3 ,1),(3, 2),(3, 3)

Totally ten times "btech" printed.

Q16

Joystick

Solution

Joysticks are often used to control video games.

Q17

All 1, 2, 3 and 4

Solution

An attachment can be a picture, Microsoft Word document, movie, sound file, Excel spreadsheet, or any other file.

Q18

1024

Solution

1MB= 1024 KB

Q19

A system designed to prevent unauthorized access

Solution

A firewall is a network security device that monitors incoming and outgoing network traffic and decides whether to allow or block specific traffic based on a defined set of security rules.

Q20

System software

Solution

An operating system (OS) is system software that manages computer hardware, software resources, and provides common services for computer programs.

Q21

1 & 2

Solution

The Excel EXACT function compares two text strings, taking into account upper and lower case characters, and returns TRUE if they are the same, and FALSE if not. EXACT is case-sensitive.

FIND- The position of a substring within a larger text string. FIND is case-sensitive.

Q22

E-mail client

Solution

Microsoft Outlook Express is a free, basic email program. You can import email messages and account information from Outlook Express into Microsoft Outlook.

Q23

Direct

Solution

The direct file organization is used to improve the performance in accessing the records as compared to sequential file organization. In direct file organization, the key value is mapped directly to the storage location.

Q24 Terabyte

Solution

Unit	Shortened	Capacity
Bit	b	1 or 0 (on or off)
Byte	B	8 bits
Kilobyte	KB	1024 bytes
Megabyte	MB	1024 kilobytes
Gigabyte	GB	1024 megabytes
Terabyte	TB	1024 gigabytes
Petabyte	PB	1024 terabytes
Exabyte	EB	1024 petabytes
Zettabyte	ZB	1024 exabytes
Yottabyte	YB	1024 zettabytes

Q25 Reply all

Solution

The reply all function means that you email your response to both the sender of the previous message and everyone who received the original email.

Q26 Signature

Solution

An email signature is text, like your contact information or a favorite quote, that's automatically added at the end of Gmail messages as a footer.

Q27 RAM

Solution

RAM is a primary storage memory while others are secondary storage memory.

Q28 All 1, 2 and 3

Solution

Signature can have text, link and image.

Q29

name@website.info

Solution

name@website.info is the correct format eg: employee@iamneo.ai

Q30

1, 2 & 3

Solution

In Incognito, none of your browsing history, cookies and site data, or information entered in forms are saved on your device. This means your activity doesn't show up in your Chrome browser history, so people who also use your device won't see your activity.